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# 📘 Assignment: Number Systems in IT and Networking

## 🧠 Overview

In Information Technology, understanding number systems is essential for working with data, memory, and networking. Two key number systems used in IT are:

- **Binary (Base-2)**: Uses only 0 and 1. It is the fundamental language of computers.
- **Hexadecimal (Base-16)**: Uses digits 0–9 and letters A–F. It is often used for compact representation of binary data.

These systems are especially important in networking, where they are used in:

- **IPv4 addresses** (e.g., `192.168.1.1`)
- **IPv6 addresses** (e.g., `2001:0db8:85a3:0000:0000:8a2e:0370:7334`)
- **MAC addresses** (e.g., `00:1A:2B:3C:4D:5E`)

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## 🔢 Binary and Hexadecimal Basics

### Binary (Base-2)
Each digit represents a power of 2. For example:
Binary: 1101 Decimal: (1×8) + (1×4) + (0×2) + (1×1) = 13

### Hexadecimal (Base-16)
Each digit represents a power of 16. For example:
Hex: 1A Decimal: (1×16) + (10×1) = 26

### Conversion Tips

- **Binary to Hex**: Group binary digits into sets of 4 (from right to left), then convert each group to hex.
- **Hex to Binary**: Convert each hex digit to its 4-bit binary equivalent.

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## 🌐 Networking Applications

### IPv4 Addresses
IPv4 addresses are typically shown in decimal, but each octet (e.g., `192`) can be converted to binary for subnetting and routing.

**Example:**
IPv4: 192.168.1.1
Binary: 11000000.10101000.00000001.00000001
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